

The diagram illustrates a transformation U between two surfaces. On the left, a genus-2 surface (a torus with a handle) is shown. The handle is represented by two intersecting loops. The surface is labeled A . This surface is transformed by U into a surface shown inside large parentheses on the right. The transformed surface is also a genus-2 surface, but the handle is now represented by a dashed red line. The surface is labeled A and k .

$$\text{Surface } A = U \left(\text{Surface } A \text{ with dashed red handle and region } k \right)$$